Design Proposal: Mach-Zehnder Interferometers with Varying Path Length Imbalance on a Silicon-on-Insulator Photonic Chip

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***Abstract*—This proposal describes the design and simulation of four unbalanced Mach-Zehnder interferometers (MZIs) on a 220 nm silicon-on-insulator platform, built from components of the SiEPIC EBeam process design kit. Each device uses a Y-branch splitter and a broadband directional coupler combiner, and the four devices differ only in the length difference between their arms, which ranges from 10.2 µm to 1.59 mm. The circuits are simulated in Lumerical INTERCONNECT using the PDK compact models. The free spectral range of each device is predicted and will be used to extract the waveguide group index once the chip is fabricated and measured.**

***Index Terms*—Mach-Zehnder interferometers, silicon photonics, integrated photonics, SOI, group index**

I. INTRODUCTION

Silicon photonics brings optical signal processing onto chips that can be made with the same fabrication infrastructure as microelectronics. Its high refractive index contrast confines light in waveguides only a few hundred nanometres wide, which makes dense, low-cost photonic circuits possible for telecommunications, sensing and computing.

The Mach-Zehnder interferometer is one of the most widely used building blocks in this field. It splits light into two paths and recombines it, so the output depends on the phase difference accumulated between the arms. That makes it the basis of modulators, switches, filters and sensors, and also a convenient test structure: when the phase difference comes only from a known difference in arm length, the measured spectrum reveals how fast light travels in the waveguide.

In this project, four MZIs with different arm length imbalances are designed in KLayout using the SiEPIC-Tools environment, simulated in Lumerical INTERCONNECT, and submitted for fabrication. The goal is to compare simulated and measured free spectral ranges (FSR) and to extract the group index of a standard strip waveguide.

II. THEORY

In each device, input light from a grating coupler is divided equally by a Y-branch. For an ideal 50/50 splitter, the intensity and field in each arm are

$$I_1 = I_2 = I_i / 2 \tag{1}$$

$$E_1 = E_2 = E_i / \sqrt{2} \tag{2}$$

The two arms have lengths L_1 and L_2, with difference $\Delta L = L_1 - L_2$. The light is recombined in a broadband directional coupler, which has two outputs. Neglecting propagation loss, the outputs are complementary:

$$I_{o1,2} = (I_i / 2)\,[1 \pm \cos(\beta \Delta L)] \tag{3}$$

where $\beta = 2\pi n_{eff}/\lambda$ is the propagation constant. The spectrum is therefore periodic, and the spacing between neighbouring maxima or minima is the free spectral range

$$\mathrm{FSR} = \lambda^2 / (n_g\, \Delta L) \tag{4}$$

where the group index accounts for waveguide dispersion,

$$n_g = n_{eff} - \lambda\, (\mathrm{d}\, n_{eff} / \mathrm{d}\lambda) \tag{5}$$

Over the 1500–1600 nm band, the effective index is well described by a second-order expansion around λ_0:

$$n_{eff}(\lambda) = n_1 - n_2(\lambda - \lambda_0) - n_3(\lambda - \lambda_0)^2 \tag{6}$$

Equation (4) shows that a larger ΔL gives a smaller FSR. Measuring the FSR of devices with known ΔL therefore gives n_g directly.

III. MODELING AND SIMULATION

A. Waveguide Parameters

All devices use a silicon strip waveguide 500 nm wide and 220 nm thick, clad in silicon dioxide, as defined in the EBeam PDK. The waveguide compact model used in INTERCONNECT describes its fundamental TE mode with an effective index $n_{eff} \approx 2.44$ and a group index $n_g \approx 4.2$ at 1550 nm, which are the values used in the calculations below.

B. Circuit Design

The circuits were laid out in KLayout with SiEPIC-Tools using the EBeam PDK: ebeam_gc_te1550 grating couplers, an ebeam_y_1550 Y-branch splitter, an ebeam_bdc_te1550 broadband directional coupler as combiner, and Waveguide cells generated with the Path-to-Waveguide tool (Fig. 1). Each MZI uses three grating couplers on a 127 µm pitch, matching the fiber array of the automated test setup: one input and one for each coupler output. The long arms of MZIs 2–4 are folded into loops so that all four devices fit in the 605 × 410 µm design area.

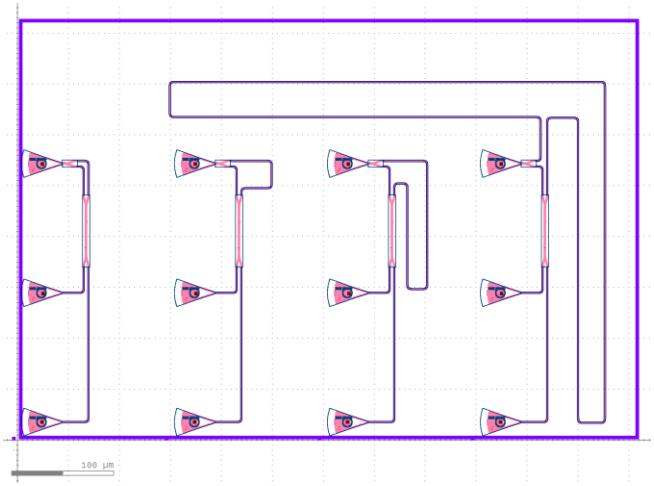


Fig. 1. KLayout layout of the four MZIs (left to right: MZI 1–4) within the 605 × 410 µm design area. Grating couplers are on the left of each device; the long arms are folded to the right.

Table I lists the arm lengths reported by SiEPIC. The short arm is 32.87 µm in every device, while the long arm is 43.068, 99.405, 309.182 and 1626.425 µm for MZIs 1–4. The predicted FSR follows from (4) with n_g = 4.2, and the simulated FSR is read from the INTERCONNECT spectra in Fig. 2.

TABLE I

Arm Lengths, Predicted and Simulated FSR

Device	ΔL (µm)	Predicted FSR (nm)	Simulated FSR (nm)
MZI 1	10.198	56.1	≈ 62
MZI 2	66.535	8.60	≈ 8.7
MZI 3	276.312	2.07	≈ 2.1
MZI 4	1593.555	0.359	not resolved

C. Circuit Simulation

The netlist of each device was exported from KLayout and simulated in Lumerical INTERCONNECT with an optical network analyzer sweeping 1500–1600 nm. The compact models include the measured wavelength response of the grating couplers, which gives the overall bell-shaped envelope peaking near 1550 nm with about 5–6 dB fiber-to-fiber insertion loss.

Fig. 2 shows the simulated spectra of the two outputs of each device. For MZI 1, minima of one output occur at 1511.8 and 1573.8 nm, an FSR of 62 nm, and the complementary output has its minimum at 1542 nm, as predicted by (3). This is about 10% larger than the 56 nm predicted from the drawn length difference, a gap that corresponds to roughly 1 µm of ΔL and is within the uncertainty of where the arm lengths begin and end inside the Y-branch and coupler.

MZI 2 shows twelve clear fringes with an average spacing of 8.7 nm, close to the predicted 8.60 nm; inserting this value into (4) gives $n_g \approx 4.13$. MZI 3 shows evenly spaced fringes about 2.1 nm apart, matching the predicted 2.07 nm. Both devices keep extinction ratios of 35–60 dB across the band. For MZI 4 the predicted FSR is only 0.359 nm, and the simulated spectrum appears as a dense band with a scalloped envelope, because the analyzer points are spaced too widely to resolve each fringe. Rerunning with at least 50,000 points over 1500–1600 nm (Appendix) is needed to resolve its minima.

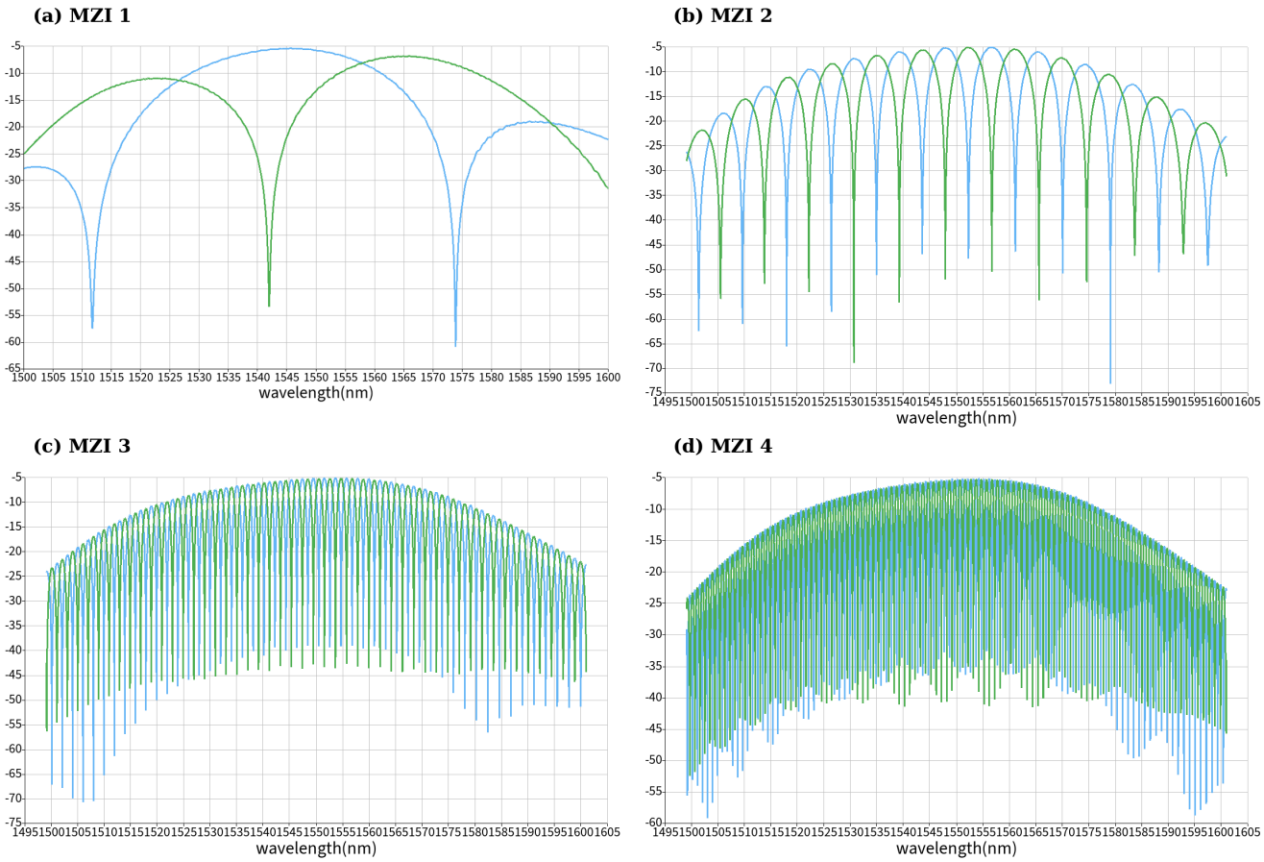


Fig. 2. INTERCONNECT simulation of (a) MZI 1, (b) MZI 2, (c) MZI 3 and (d) MZI 4: transmission (dB) of the two outputs of the directional coupler from 1500 to 1600 nm. The FSR decreases as ΔL increases; the MZI 4 spectrum is under-sampled.

IV. FABRICATION

The layout will be fabricated using 100 keV electron-beam lithography on a silicon-on-insulator wafer with a 220 nm silicon device layer on a 3 µm buried oxide, followed by fully-etched reactive ion etching and oxide cladding. The four devices occupy the allocated 605 × 410 µm area (Fig. 1). Each input grating coupler carries an opt_in text label so that the automated test system can locate and identify the device.

V. EXPERIMENT

The chip will be measured on an automated probe station with a tunable laser swept from 1500 to 1600 nm in fine steps, coupling light in and out through a polarization-maintaining fiber array aligned to the grating couplers for the TE mode. The fine step is needed to resolve the 0.36 nm FSR of MZI 4.

VI. RESULTS AND DISCUSSION

The measured spectra will be corrected for the grating coupler envelope, the minima will be located, and the FSR versus wavelength will be extracted for each device. The group index will then be obtained from (4) and compared with the simulated value of 4.20. Differences are expected from fabrication variation in waveguide width and thickness, which shifts the fringe positions but changes the group index only slightly.

VII. CONCLUSION

Four unbalanced MZIs with path differences spanning more than two orders of magnitude have been designed so that the same chip shows both widely and closely spaced interference fringes. Together they will allow an accurate measurement of the waveguide group index and a direct comparison between simulation and fabricated devices.

REFERENCES

[1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems*. Cambridge University Press, 2015.

[2] SiEPIC EBeam Process Design Kit and SiEPIC-Tools, github.com/SiEPIC.

[3] Ansys Lumerical INTERCONNECT documentation, optics.ansys.com.